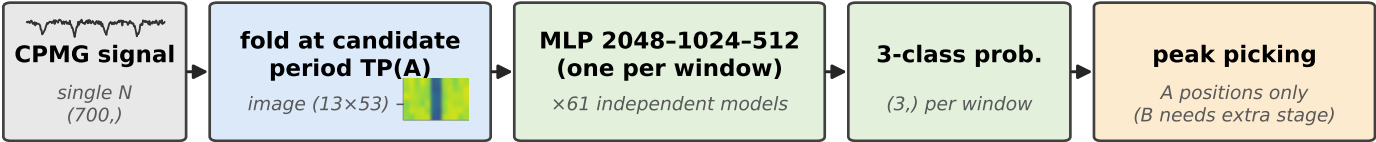


(a) Jung et al. (2021) — window-bank MLP

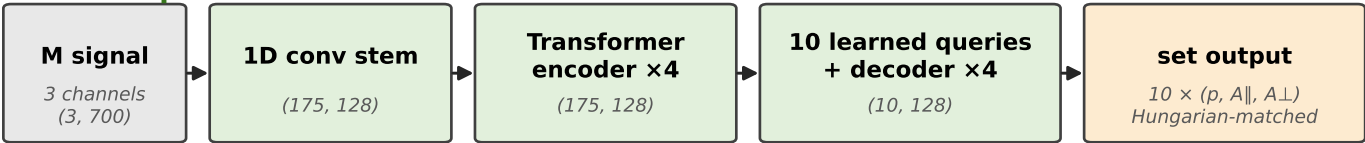
reproduced baseline



no information shared across windows · separate denoiser harms narrow dips at room temperature

(b) SpinDETR — end-to-end set prediction

no physics prior (ablation)



direct (A, B) regression in one forward pass · learns periodicity from data alone

(c) PeriodFormer (ours) — window tokens + attention

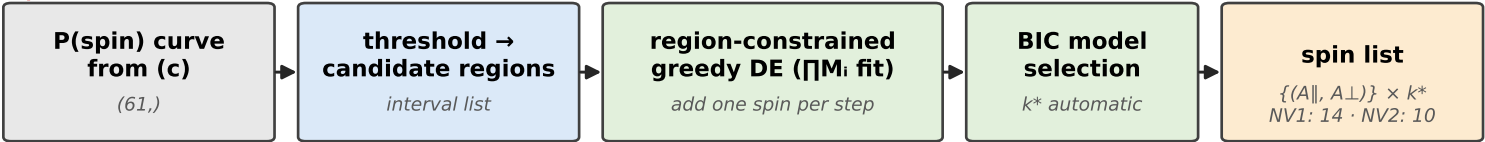
physics prior + attention



keeps the period-folding prior of (a), replaces 61 independent classifiers with ONE model attending across windows → zero false positives in all benchmarks

(d) PF→DE hybrid (ours, final)

detect × enumerate



multiple spins allowed per region (cluster enumeration) · best F1 in every ground-truth benchmark arm (cryo / room-T / model-mismatch)

□ input □ representation □ computation □ output